Ocean Best Practices System (OBP-S)

IODE OceanBestPractices (OBP-R): enhancements and future work

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ETDMP-VI
6th Session JCOMM/IODE Expert Team on Data Management Practices
17 - 19 Sep 2018, Oostende, Belgium

OceanBestPractices – background

- **2014** – **OceanDataPractices,** an open access repository, created as a deliverable of the IODE Ocean Data Standards (and Best Practices Project).
  Replaced *JCOMM Catalogue of Best Practices* (not a searchable database and records not validated) – 163 records migrated to ODP as initial content.
  2015 - Records reviewed by Mathieu Ouellet.

- **2016 –** ODIP and AtlantOS Projects Work Packages on Best Practices looking to setup a permanent archive of ocean best practices to serve the ocean observing community – IODE(PP) offered ODP for this purpose. IODE became a partner in the AtlantOS Best Practices Working Group.

- **2017 –** Agreement by main ODP partners **(IODE/JCOMM/WMO/ICES)** to change the name to **OceanBestPractices** to reflect the broader remit of 'all ocean-related' best practices.

- **2018/9 –** Agreement for OBP-R(S) to become an IODE Project with its own SG governance
  - Proposal to IODE MG/Committee
  - Approval as a formal IODE Project

AtlantOS Best Practices Working Group

Best Practices Working Group:

Mark Bushnell	IOOS
Pier Luigi Buttigieg	AWI
Juliet Hermes	SAEON/JCOMM
Emma Heslop	GOOS/JCOMM
Frank Muller-Karger	IMaRS, Un S Florida
Johannes Karstensen	GEOMAR
Cristian Muñoz	SOCIB
Francoise Pearlman	IEEE
Jay Pearlman **(LEAD)**	IEEE
Pauline Simpson	IODE

Best Practices Working Group:

Mark Bushnell	IOOS
Pier Luigi Butti	
Juliet H	COMM
Emma	OMM
Frank M	n S Florida
Johanne	
Cristian	SOCIB
Francoise Pearlman	IEEE
Jay Pearlman **(LEAD)**	IEEE
Pauline Simpson	IODE

Funded by:
- AtlantOS
- ODIP
- NSF RCN

Challenges of Best Practices Documents

Large investments in obtaining high quality ocean observing measurements.

For quality control and intercomparison it is important to document methods and procedures

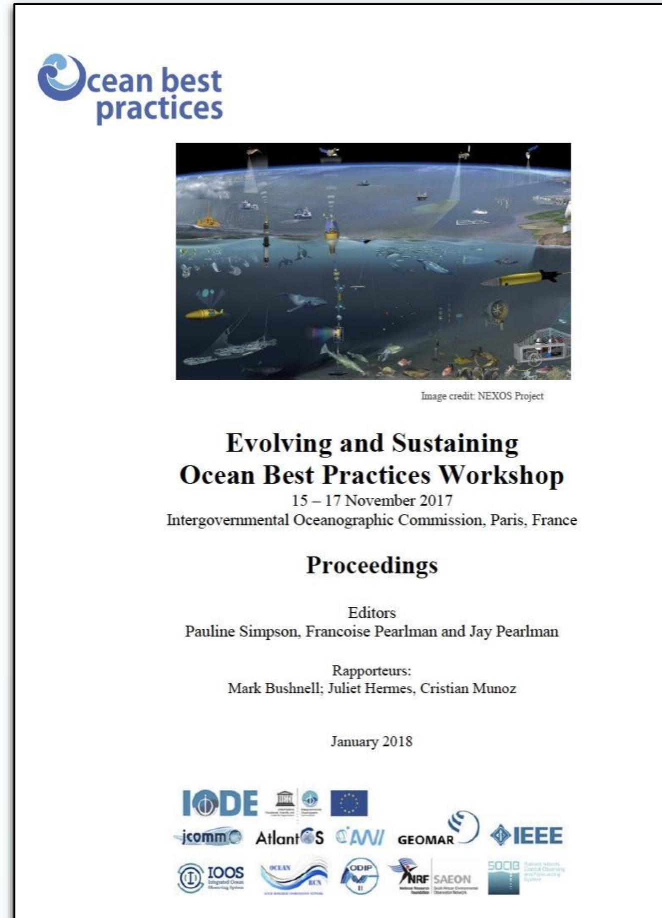
But:

- Quality of BP Documentation varies widely
- Data and metadata formats are inconsistent
- Machine readability is limited (if at all)
- Sustainability is often not guaranteed

Best Practice Definition

"A community best practice is a methodology that has repeatedly produced superior results relative to other methodologies with the same objective.

To be fully elevated to a best practice, a promising method needs to be adopted and employed by multiple organizations."



Ocean best practices

Image credit: NEXOS Project

Evolving and Sustaining Ocean Best Practices Workshop

15 – 17 November 2017

Intergovernmental Oceanographic Commission, Paris, France

Proceedings

Editors

Pauline Simpson, Francoise Pearlman and Jay Pearlman

Rapporteurs:

Mark Bushnell; Juliet Hermes, Cristian Munoz

January 2018

AtlantOS Best Practices Working Group

- **Vision:** Increase efficiency, reproducibility and interoperability of the entire ocean observing value chain by providing the community with a unified, sustained and readily accessible knowledge base of interdisciplinary best practices
- **Mission**: *to provide coordinated and sustained global access to best practices in ocean observing to foster innovation and excellence by developing a system and engaging ocean observing communities in a joint and coordinated effort in producing, reviewing and sustaining BP documents*.

Elements of the OBP System

Participating Organizations and Programs

Peer Review Journal

Repository

Advanced Technology

Support
Training, etc

The User

Elements of the OBP System

Participating Organizations and Programs

BP documents and peer review articles deposited into the UNESCO/IOC IODE best practice repository

Peer Review Journal

Repository

IODE

Advanced Technology

Support Training, etc

The User

Elements of the OBP System

Participating Organizations and Programs

BP creators depositing into the UNESCO/IOC IODE best practice repository
Documents and peer review journal articles on BP

Peer Review Journal

Repository

Support Training, etc

The User

Advanced Technology: Semantic indexing and natural language processing to increase discoverability. Text-mining technologies used to link all best practice documents in the system with fusion of ocean ontologies

Elements of the OBP System

Participating Organizations and Programs

BP creators depositing into the UNESCO/IOC IODE best practice repository
Documents and peer review journal articles on BP

Advanceed Technology: Semantic indexing and natural language processing to increase discoverability. Text-mining technologies used to link all best practice documents in the system with fusion of ocean ontologies

Peer Review Journal

Internationally recognized open access repository

Advanced Technology

Support Training, etc

OTGA, POGO etc
Training Course :
Online and Blended

The User

Elements of the OBP System

Participating Organizations and Programs

BP creators depositing into the UNESCO/IOC IODE best practice repository Documents and peer review journal articles on BP

Advanced Technology: Semantic indexing and natural language processing to increase discoverability. Text-mining technologies used to link all best practice documents in the system with fusion of ocean ontologies

Peer Review Journal

Repository

Advanced Technology

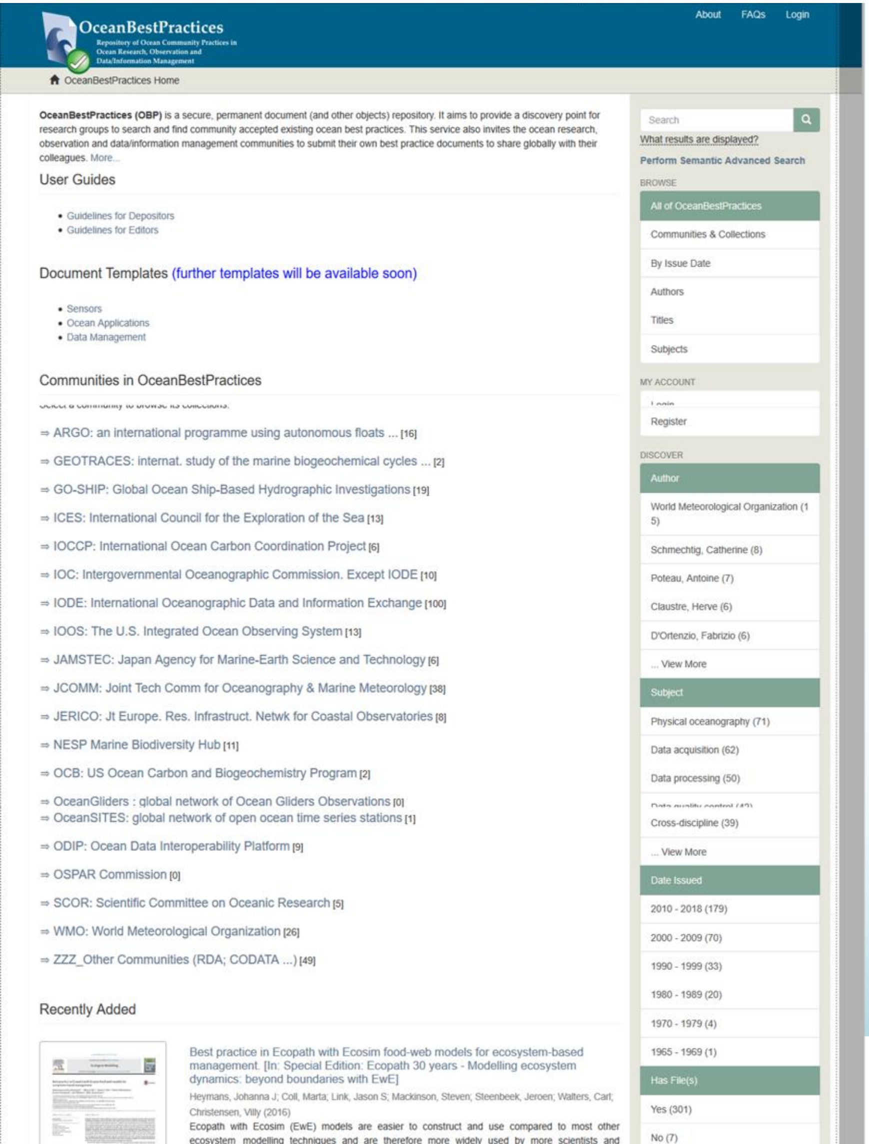
OTGA, POGO etc Training Courses : Online and Blended

Support Training, etc

The User

OceanBestPractices Repository

https://www.oceanbestpractices.net



- **The HUB of the Ocean Best Practices System**
- A central, permanent, curated, open access repository.
- Provenance of UNESCO/IOC – IODE encourages BP providers to trust its permanence and sustainability.
- Submissions will be FAIRified in the OBP-R

A FAIRer future for Ocean BPs

TO BE FINDABLE:
F1. (meta)data are assigned a globally unique and eternally persistent identifier.

F2. data are described with rich metadata.

F3. (meta)data are registered or indexed in a searchable resource.

F4. metadata specify the data identifier.

TO BE ACCESSIBLE:
A1 (meta)data are retrievable by their identifier using a standardized communications protocol.

A1.1 the protocol is open, free, and universally implementable.

A1.2 the protocol allows for an authentication and authorization procedure, where necessary.

A2 metadata are accessible, even when the data are no longer available.

TO BE INTEROPERABLE:
I1. (meta)data use a formal, accessible, shared, and broadly **applicable** language for knowledge representation.

I2. (meta)data use vocabularies that follow FAIR principles.

I3. (meta)data include qualified references to other (meta)data.

TO BE RE-USABLE:
R1. meta(data) have a plurality of accurate and relevant attributes.

R1.1. (meta)data are released with a clear and accessible data usage license.

R1.2. (meta)data are associated with their provenance.

R1.3. (meta)data meet domain-relevant community standards.

System offers End to End Management

1. Support for BP Creation and Deposit

- Want to minimize the effort involved in creating, depositing and updating best practices.

- The ideal solution is to have the repository populated by automated ingest of BP documents created using agreed templates (for machine readability).

  - ✓ BP Document (Meta)Data Sheet (reflects the enhanced Repository metadata profile)
  - ✓ BP Document Template (sensors, applications, data management, but more on the way in collaboration with community)

Document Data Sheet v4.2 (for submissions to www.oceanbestpractices.net)

We recommend including this document data sheet into your Best Practice document. **Please do not change any formatting, entries in the left column, or the table structure.** The format below will allow automatic ingest of the data in this table into the OceanBestPractices Repository. Enter data only in the right-hand column.

Mandatory fields are indicated with ** but we strongly recommend that you provide data (if applicable) for all the metadata fields requested; this will allow you to unambiguously declare what your best practice is about and help our indexing technology make it more visible.

Best practice type Choose upto 2 entries from the list below. Separate multiple entries with a semicolon (;) • Best Practice • Standard Operating Procedure • Manual (incl. handbook; guide, cookbook)	
English-language document title ** Entries should be in English. If applicable, include a sub-title after a colon (:) and version code after the text (e.g. Version 3.2).	
Original document title If the title was not originally in English, please include it in its original form here. If applicable, include a sub-title after a colon (:) and version	

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Original document title If the title was not originally in English, please include it in its original form here. If applicable, include a sub-title after a colon (:) and version	

Essential Ocean Variables (EOVs) ** Select names (e.g. sea ice) from this link Separate multiple entries with a semicolon (;).	
Sustainable Development Goals and Targets eg. 14.1;15.3 SDG Target codes are of the form [Goal number].[Target number]. Refer to this page for more information: https://sustainabledevelopment.un.org/ Separate multiple entries with a semicolon (;)	

Maturity Level If applicable, note the maturity level of this BP, following the guidelines set forth here: https://esto.nasa.gov/files/trl_definitions.pdf	

Maturity Level (Technology Readiness Levels)

[based on NASA ESTO TRL Definitions https://esto.nasa.gov/files/trl_definitions.pdf]

1	Basic principles of technology observed & reported
2	Technology Concept and/or Application Formulated
3	Analytical and Laboratory Studies to validate analytical predictions
4	Component and/or basic sub-system technology valid in lab environment
5	Component and/or basic sub-system technology valid in relevant environment
6	System/sub-system technology model or prototype demo in relevant environment
7	System technology prototype demo in an operational environment
8	System technology qualified through test & demonstration
9	System technology 'qualified' through successful mission operations

BP Template

TABLE OF CONTENTS

Best practices may be documented in the form of standard operating procedures or manuals or other formats. The list below are suggested sections and not all of these may be occurring in the document. Replace the page number with N/A if the section is not applicable. Additional topic titles should be added to the table of contents in the appropriate section, inserting sequential number

Sensors
Applications
Data Management

More to come

Creating a weekly Harmful Algal Bloom bulletin. Version 1.0. [Best Practice Description Document]

This document describes the procedural steps in creating an information product focused on toxic and harmful phytoplankton. The product is an online Harmful Algal Bloom (HAB) bulletin for aquaculturists, who can face serious operational challenges in the days after a HAB event. Data from satellite, numerical hydrodynamic models and In-situ ocean observations are organised and presented into visual information products. These products are enhanced through local expert evaluation and their interpretation is summarised in the bulletin. This document aims to provide both process overviews (the "what" of the Best Practice in producing the bulletins) and detail procedures (the "how" of the Best Practice") so that the bulletins may be replicated in other geographic regions......

View/Open

PDF (3.656Mb)

Date

Resource URL

https://oar.marine.ie/handle/10793/1344

Publisher

Marine Institute
Galway, Ireland

Document Language

en

Search

Search OceanBestPractices
This Collection

What results are displayed?

Perform Semantic Advanced Search

BROWSE

All of OceanBestPractices
Communities & Collections
By Issue Date
Authors
Titles
Subjects

This Collection
By Issue Date

System offers End to End Management
2. Encourage BP Creation/Creators

- Offering a route to Academic recognition through peer reviewed article on a BP
- Introduction of a new peer review journal
- **Frontiers in Marine Science**
  (Research Topic: Best Practices in Ocean Observing)
  http://journal.frontiersin.org/researchtopic/7173).

  ✓ Acceptance of article by journal indicates the Best Practice document peer review and gives it a “seal of quality

  ✓ To publish in journal – requirement (where appropriate) to deposit the BP in OBP-R

Frontiers in Marine Science
(Research Topic: Best Practices in Ocean Observing)

https://www.frontiersin.org/research-topics/7173/best-practices-in-ocean-observing

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Ocean best practices

Research Topic

Best Practices in Ocean Observing

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VIEWS
5,437

Overview | 3 Articles | 94 Authors | Impact | Comments

About this Research Topic

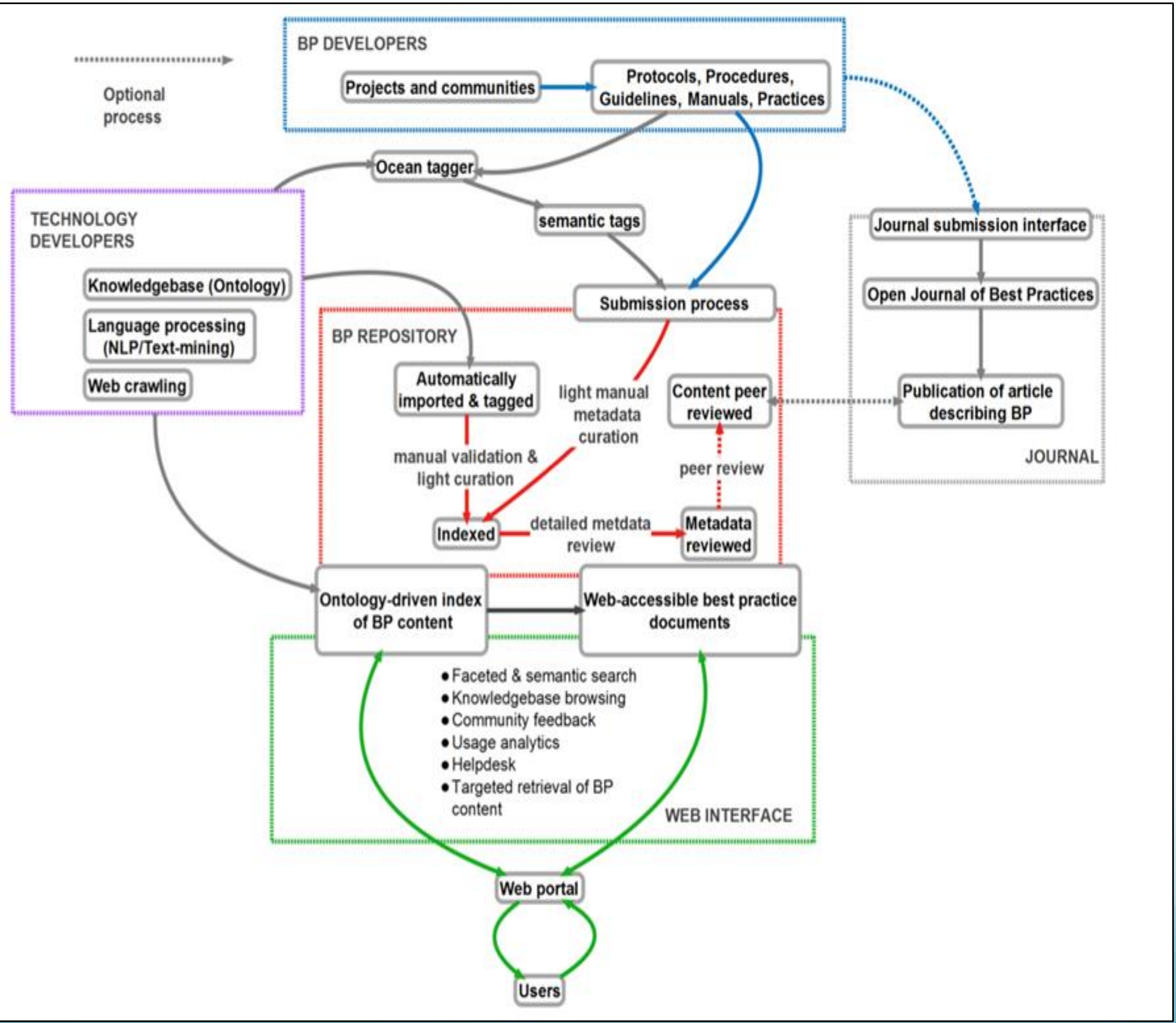
An immense corpus of well-tested methodology in ocean observing has been and is being amassed by national, regional, and international observatory networks. However, despite the quality of these efforts, the discoverability and sustainability of high-quality methodology is still limited by fragmented ...

Topic Editors

Johannes Karstensen
GEOMAR Helmholtz-Zentrum für Ozeanforschung

Follow

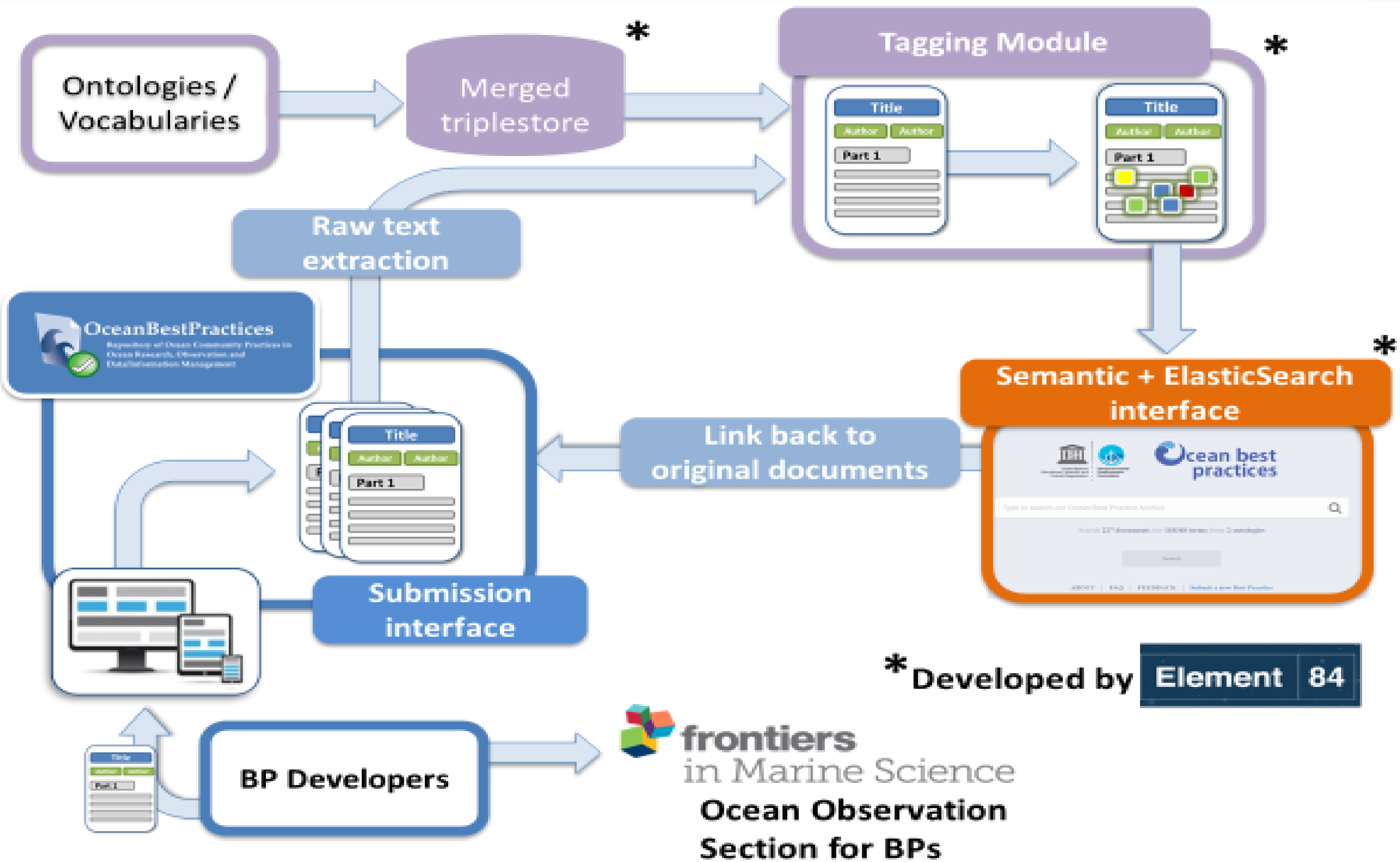
System offers End to End Management
3. OceanBestPractices Architecture



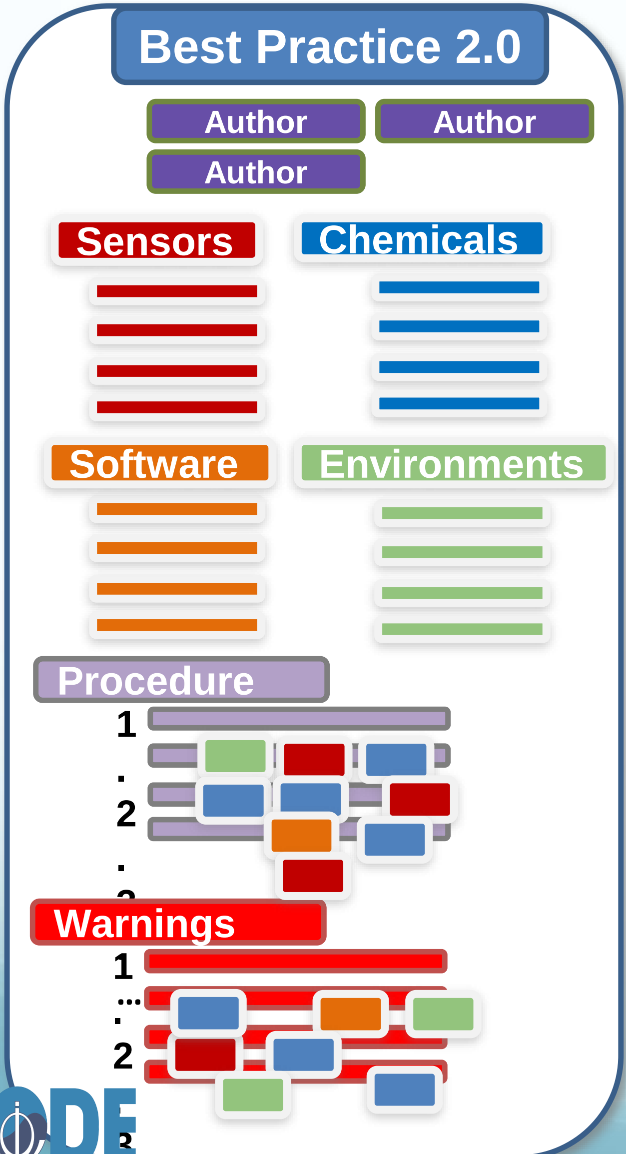
The main components of the Ocean Best Practices System revolve around the permanent repository hub:

- BP Creation by the ocean observing community Submission by BP practitioners and by web crawling algorithms and
- Frontiers in Marine Research RT; mandatory that the BP is already uploaded to the repository
- Semantic tagging using a fusion of ocean-related ontologies
- Curation of BP metadata and file with DOI issued. Review by GOOS EOV Panel
- Community facing User Interface with sophisticated semantic and natural language query and reporting options

Advanced Technology
Semantic Indexing and Ontologies



OceanKnowledge Tagger Module

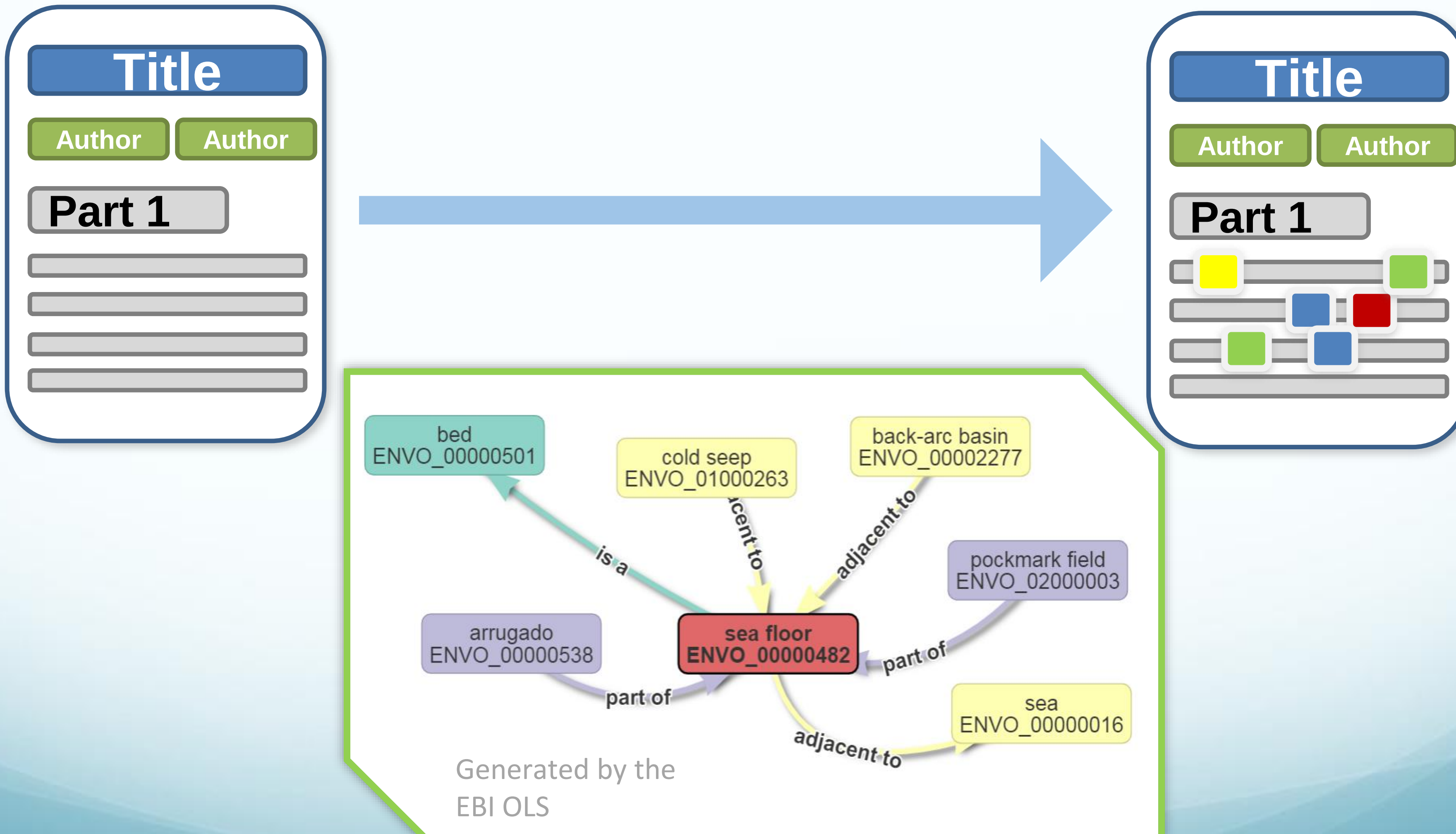


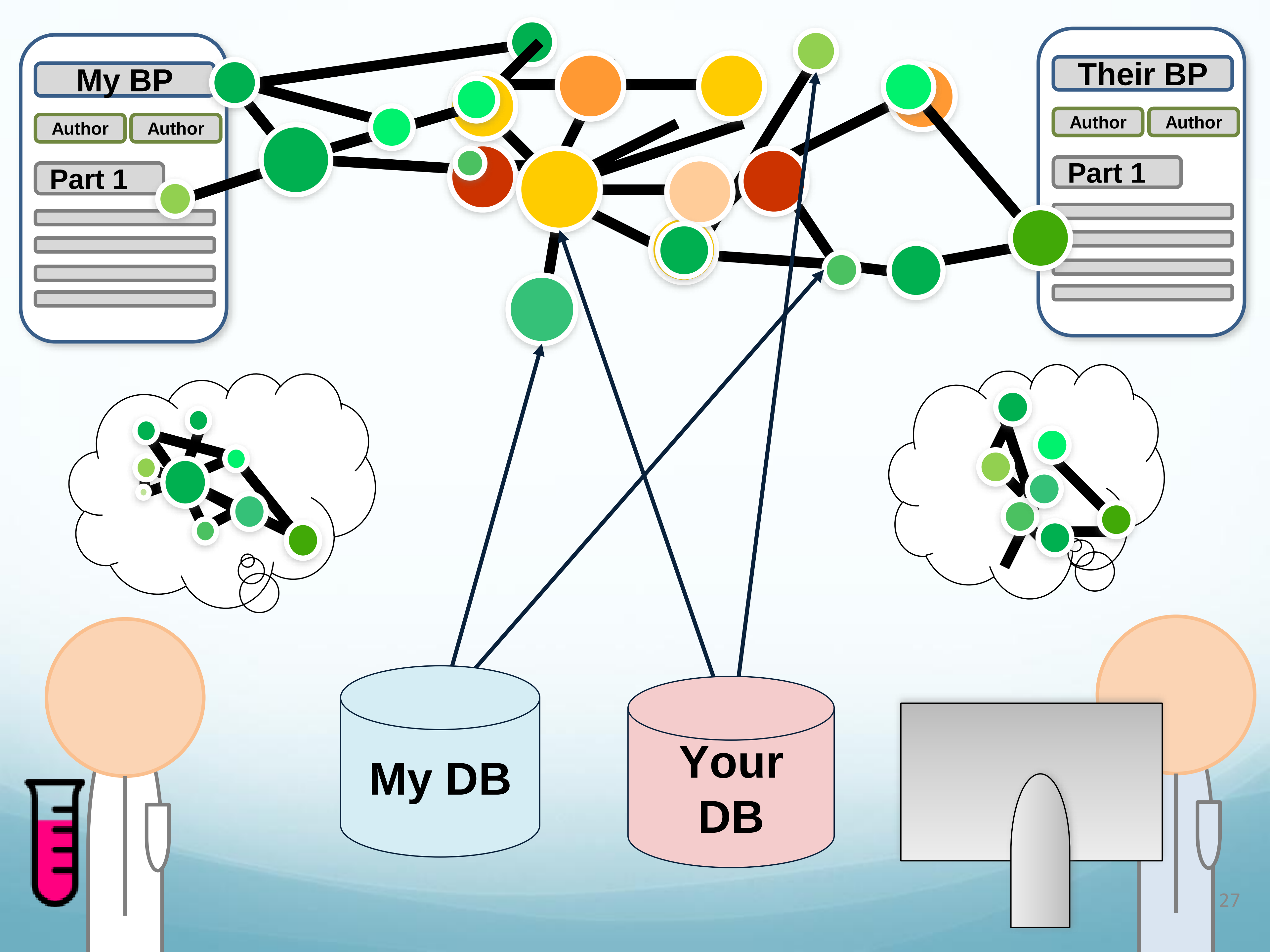
Knowledge-linked best practices:

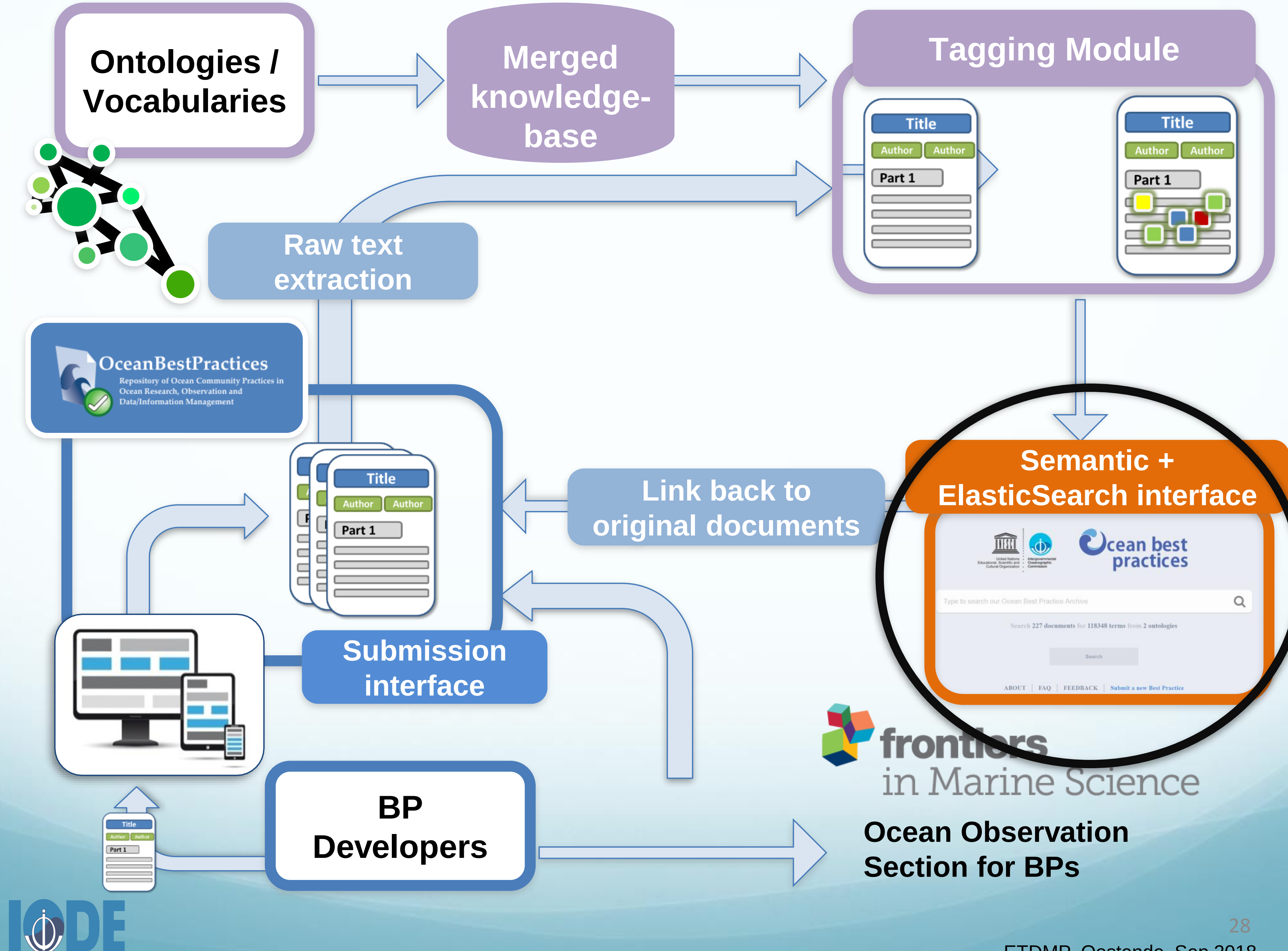
Using Semantics and Natural Language Processing in Best Practice documents

- Retrieve the common chemicals used in "Best practices for nitrate measurements" and "Best practices for nitrous oxide measurements"
- Retrieve all the sensors manufactured by company X used to sense oxygen in *any* deep sea environment
- What software does the author Jane Doe use in her best practices about coral reefs?

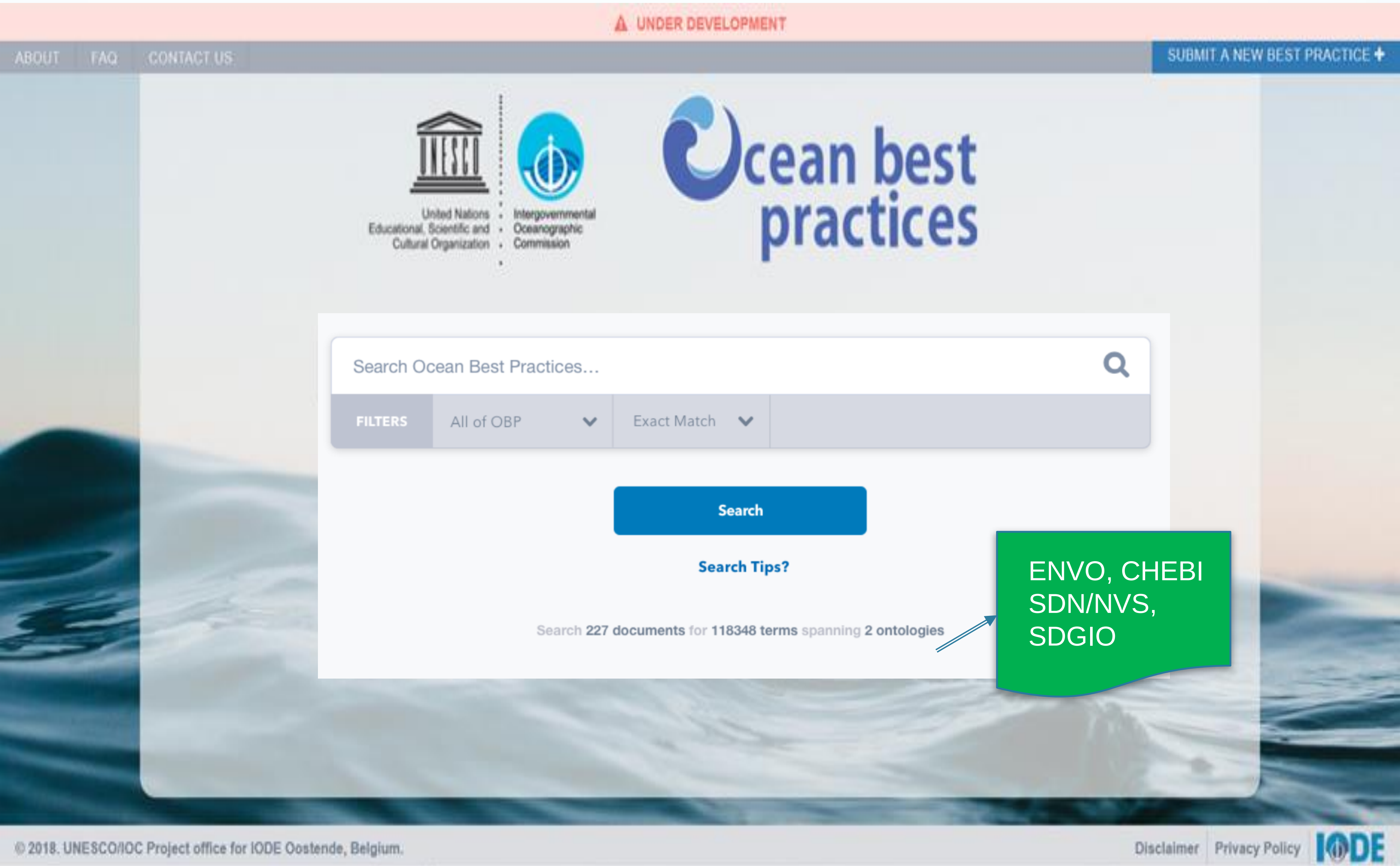
This document includes methods to sense seepage of methane from the ***seafloor*** …



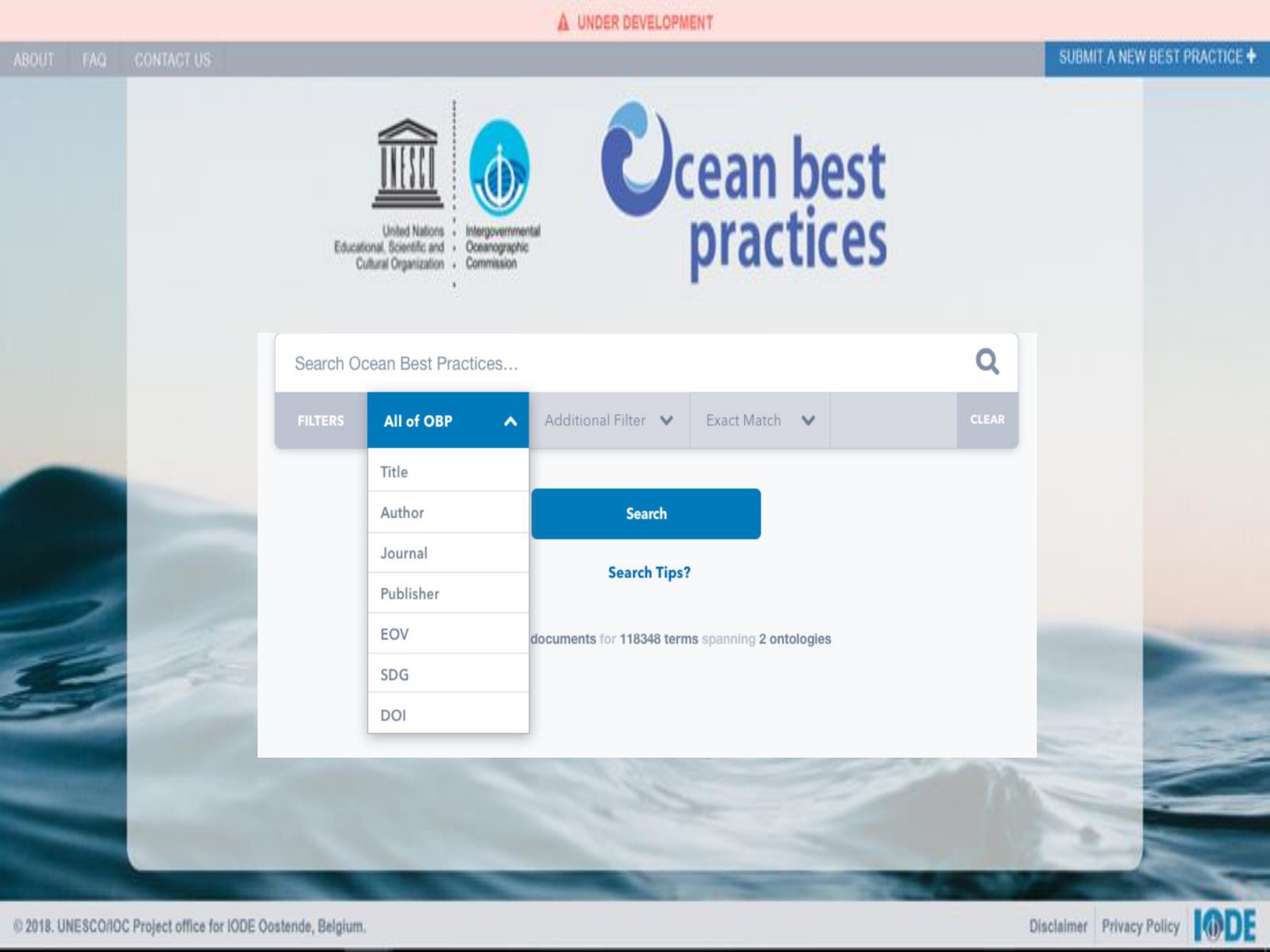

My BP
Author
Author
Part 1
Their BP
Author
Author
Part 1
My DB
Your DB


Ontologies / Vocabularies
Merged knowledge-base
Tagging Module
Title
Author
Author
Part 1
Title
Author
Author
Part 1
Raw text extraction
OceanBestPractices
Repository of Ocean Community Practices in Ocean Research, Observation and Data/Information Management
Title
Author
Author
Part 1
Link back to original documents
Semantic + ElasticSearch interface
Ocean best practices
Type to search our Ocean Best Practice Archive
Search 227 documents for 118348 terms from 2 ontologies
Search
ABOUT | FAQ | FEEDBACK | Submit a new Best Practice
Submission interface
frontiers in Marine Science
Ocean Observation Section for BPs
BP Developers
Title
Author
Author
Part 1
IODE

oceanbestpractices.org – new UI



ENVO, CHEBI
SDN/NVS,
SDGIO


UNDER DEVELOPMENT
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SUBMIT A NEW BEST PRACTICE
United Nations Educational, Scientific and Cultural Organization
Intergovernmental Oceanographic Commission
Ocean best practices
Search Ocean Best Practices...
FILTERS
All of OBP
Additional Filter
Exact Match
CLEAR
Title
Author
Journal
Publisher
EOV
SDG
DOI
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documents for 118348 terms spanning 2 ontologies
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United Nations Educational, Scientific and Cultural Organization

Intergovernmental Oceanographic Commission

Ocean best practices

nitra

- nitracrine
- nitrate ester
- nitrates
- nitrazepam
- nitrate salt
- nitrate
- nitramine

UNDER DEVELOPMENT

Ocean best practices

nitrate

ORGANIZATIONS THAT SUPPORT THIS EFFORT | IODE | BPWG

chlorophyll a (CHEBI)

marine environment (ENVO)

carbon dioxide (CHEBI)

ocean acidification (ENVO)

research station (ENVO)

m-cresol (CHEBI)

fossil fuel (CHEBI)

climate change (ENVO)

sea water (ENVO)

ALL TAGS *(161)*

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36 results Sort By YEAR (desc)

2017 en

Report on the status of sensors used for measuring nutrients, biology-related optical properties, variables of the marine carbonate system, and for coastal profiling, within the JERICO network and, more generally, in the European context.

Reagents and the on board ***nitrate*** standard are placed in bags within the analyser housing.,The ***nitrate*** method applied on the CHEMINI is based on the Griess assay except for the reduction of ***nitrate***,To ensure the long term stability of the on board ***nitrate*** standard on the NAS, Cefas autoclave the ***nitrate***,of the traditional copperized cadmium column, for the reduction of ***nitrate*** to nitrite.,For the analysis of ***nitrate***, the µMAC-1000 uses a copperized Cd-column for reduction of ***nitrate*** to nitrite

thumbnail

x Viewing Tags

Ifremer for JERICO-NEXT Project

Explore knowledge neighbourhood to enhance discovery

2017 en

A User's guide for selected autonomous biogeochemical sensors. An outcome from the 1st IOCCP International Sensors Summer Course, June 22 – July 1, 2015, Kristineberg, Sweden.

thumbnail

IODE

UNDER DEVELOPMENT

Ocean best practices

nitrate

ORGANIZATIONS THAT SUPPORT THIS EFFORT IODE | BPWG

chlorophyll a (CHEBI)

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Link to the section in the document

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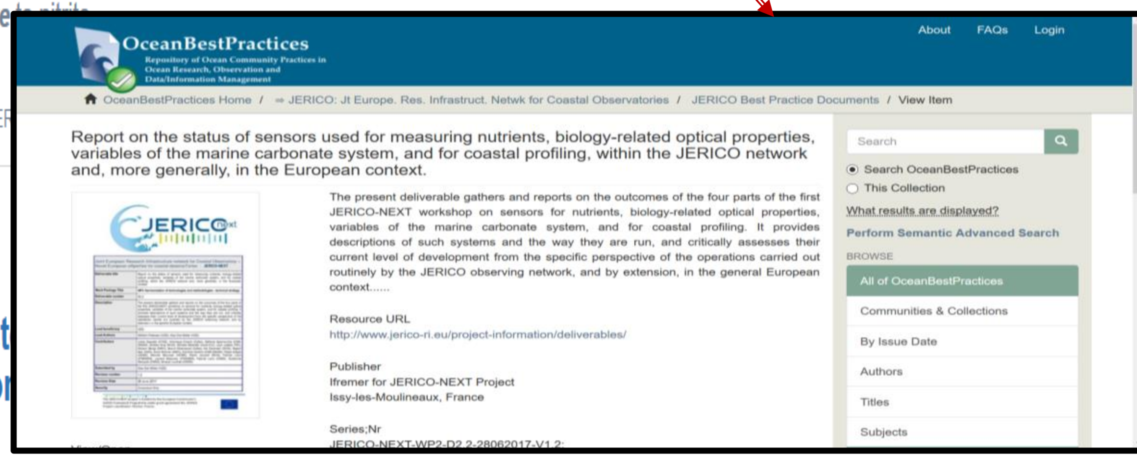

OceanBestPractices
Repository of Ocean Community Practices in Ocean Research, Observation and Data/Information Management

About FAQs Login

OceanBestPractices Home / ⇒ JERICO: Jt Europe. Res. Infrastruct. Netwk for Coastal Observatories / JERICO Best Practice Documents / View Item

Report on the status of sensors used for measuring nutrients, biology-related optical properties, variables of the marine carbonate system, and for coastal profiling, within the JERICO network and, more generally, in the European context.

The present deliverable gathers and reports on the outcomes of the four parts of the first JERICO-NEXT workshop on sensors for nutrients, biology-related optical properties, variables of the marine carbonate system, and for coastal profiling. It provides descriptions of such systems and the way they are run, and critically assesses their current level of development from the specific perspective of the operations carried out routinely by the JERICO observing network, and by extension, in the general European context......

Resource URL
http://www.jerico-ri.eu/project-information/deliverables/

Publisher
Ifremer for JERICO-NEXT Project
Issy-les-Moulineaux, France

Series;Nr
JERICO-NEXT-WP2-D2.2-28062017-V1.2;

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All of OceanBestPractices
Communities & Collections
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Authors
Titles
Subjects

2017 en

A User's guide for select the 1st IOCCP Internatio Kristineberg, Sweden.

thumbnail

Additionally, users will be able to refine their search to only Peer Reviewed (Refereed) Best Practices:

Peer Review recorded :

- ✓ Best practices that have gone through a peer review process within their community
- ✓ Article on BP published in peer review journal indicates the review experts give a 'seal of quality' to the Best Practice document.

Yet to be implemented:

- Community review within OBP-R (feedback mechanism; 'Like' possibilities)
- GOOS Panels and their experts, as well as the journal reviewers, may serve the function of defining a recommended best practice. This will make it easier for community users to identify **the most relevant practice for their use.**
- ODSBP? – community review process for proposed standards

New functionality and Enhancements for all the elements of OceanBestPractices

Participating Organizations and Programs

Peer Review Journal

OBP Repository

Advanced Technology

Support Training, etc

The User

Achievements

BP Creation and Deposit

- ✓ BP Document Templates and Document Data Sheets (to assist machine readability and populate repository metadata fields)
- ✓ New peer review journal: *Frontiers in Marine Science, Research Topic – Best Practices in Ocean Observing* (mandatory OBP-R deposit)

Smart Repository

- ✓ New modern User Interface with sophisticated semantic search and reporting options; statistics on downloads and locations
- ✓ OBP Repository now issues DOI
- ✓ Additional non-bibliographic metadata fields: BP Type; EOV; SDG, Maturity Level; Next Review Date; BP contact email etc
- ✓ Text-mining technologies used to link all best practice documents in the system with ontologies focused on environmental phenomena, chemicals, and the SDGs - the first ontologies to be loaded are ENVO, CHEBI, SDN/NVS, SDGIO.

Outreach

- ✓ Monthly Newsletter – *Good Better Best* - email to sign up = obpcommunity@oceanbestpractices.org
- ✓ Paris Workshop, Webinars, Conference and journal contributions

Coming Soon!

- Automated Web crawling exercise
- Automated ingest of BP document and DDS metadata
- Automated email reminder to BP creator of review date
- Natural language queries
- Upload of additional ontologies
- Linked data
- Peer Review – recommended best practices; review and endorsement of BP by GOOS Expert Panels
- BP training courses – IODE OceanTeacher Global Academy
- OceanObs19 – community white paper (BPWG Jay Pearlman lead)
- SCOR WG (BPWG proposal submitted)
- Recruit Associate WG members responsible for advocacy in specific ocean disciplines

+ user requested functionality – feedback mechanism , beta test of UI
OBP is system but also a tool
e.g. Gap analysis for EOV BPs
Creation of a BP document on the OBP-R interface (a la Wikipedia)

IODE

OBP Policy Snippets

Who decides whether an item is a Best Practice?

It is the responsibility of the submitting Partner (project, organization) to define a certain item as a “Best Practice”. BP can be in form of Standard Operating Procedures, Manual, Handbook etc. It is recommended for each partner to have a well-defined/documented process for the creation and documentation of best practices.

Quality control:

the OBP-R (repository) editor carries out a check of submitted content by verifying if the full text file and all metadata fields have been completed. The editor will NOT validate the scientific content of the BP as this will be the responsibility of the BP provider.

Copyright of content:

The Best Practice creator retains ownership(copyright) of the document and is entirely responsible for the scientific content and the copyright status - The Deposit Agreement states *" I declare that I have the authority of the authors to make this agreement and I understand that IODE does not assume any responsibility for any content errors, or any breach of copyright in distributing these files or metadata. (All authors are urged to prominently assert their copyright on the title page of their work)".*

License:

OceanBestPractices recommends allocating a Creative Commons License to BP - this leaves users in no doubt as how the Best Practice can be re-used.

Community engagement is essential:
create, access, review/sustain and use

questions?